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# 9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ml4e-connectedness`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

# A Robustness

**Table 3.** Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

| Scenario | Windows | DY | Lasso VAR | NETS | FNETS (idio.) | Lost w/o factor (%) | Common var. (%) |
|---|---|---|---|---|---|---|---|
| Baseline (w=104, H=4) | 179 | 81.0 | 80.0 | 61.5 | 47.2 | 41 | 58 |
| H=2 | 179 | 80.9 | 80.0 | 61.5 | 47.2 | 41 | 58 |
| H=8 | 179 | 81.0 | 80.1 | 61.5 | 47.2 | 41 | 58 |
| w=78 | 185 | 81.2 | 79.6 | 57.8 | 50.5 | 38 | 58 |
| w=130 | 172 | 81.1 | 80.5 | 63.4 | 46.2 | 43 | 58 |
| Excluding SPY | 179 | 80.0 | 78.9 | 59.9 | 45.9 | 42 | 58 |
| Excluding windows touching 2020 | 141 | 79.7 | 78.6 | 58.8 | 47.2 | 41 | 55 |

**Table 4.** Static connectedness: one estimate on the whole period (full sample) and on the subsamples before and after 2020. These are single long-sample estimates, not averages over rolling windows, so they differ from Tables 2 and 3.

| Sample | Method | Total conn. (%) | LR p.corr. dens. (%) |
|---|---|---|---|
| Full sample | DY | 81.9 | 100.0 |
| Full sample | Lasso VAR | 82.1 | 100.0 |
| Full sample | NETS | 73.5 | 60.1 |
| Full sample | FNETS (idio.) | 34.2 | 0.0 |
| 2011-2019 | DY | 83.0 | 100.0 |
| 2011-2019 | Lasso VAR | 82.8 | 100.0 |
| 2011-2019 | NETS | 69.6 | 60.1 |
| 2011-2019 | FNETS (idio.) | 34.1 | 2.0 |
| 2021-2026 | DY | 73.5 | 100.0 |
| 2021-2026 | Lasso VAR | 73.7 | 100.0 |
| 2021-2026 | NETS | 56.0 | 52.3 |
| 2021-2026 | FNETS (idio.) | 39.6 | 13.7 |

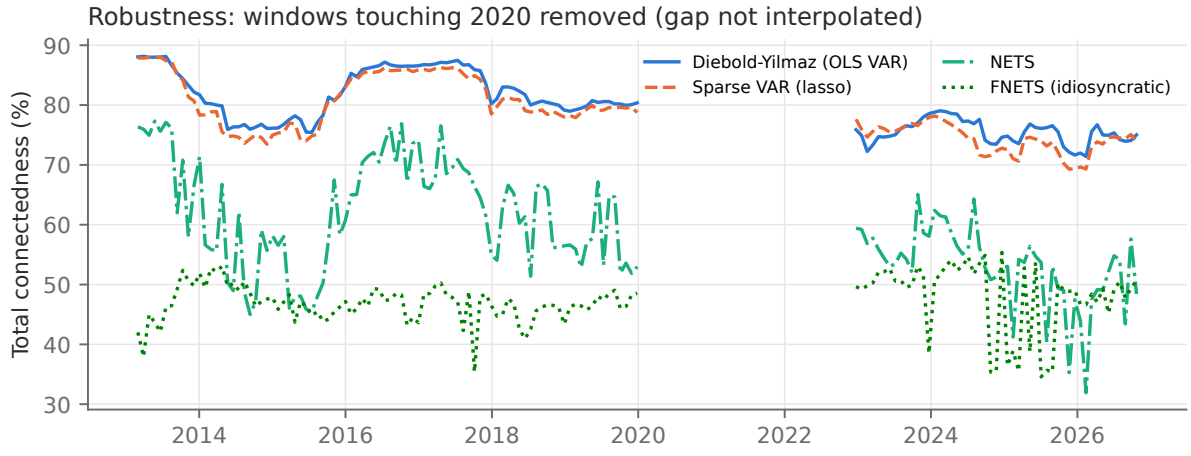



**Fig. 6.** Total connectedness excluding every window that contains a week of 2020.

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5%